SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602 105

CO2 Problem Solving Assignment: Standards-Compliant Laser Transmitter for an 8-Channel DWDM Link

ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

Submitted by

Field	Details
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Programme	B.E. Electronics and Communication Engineering
Course	ECA1804 – Optoelectronics and Optical Communication
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Overview and given data

Answer in one line: an externally modulated (zero-chirp MZM), TEC-stabilised 1550 nm InGaAsP/InP MQW DFB laser array, monolithically integrated on InP with an AWG multiplexer, launched at 2 mW/channel with SBS phase dithering and per-span dispersion management, meets the 8 × 25 Gb/s, 6 × 80 km DWDM requirement. Soliton transmission (N = 0.17) and digital backpropagation are not appropriate for this link.

Field	Detail
Course	ECA18 – Optoelectronics and Optical Communication
Assessment	CO2 Problem Solving Assignment (100 marks), K3 Apply / K4 Analyze
Student name / Reg. No.	_____
Title	Standards-compliant laser transmitter design, wavelength control and nonlinearity management for an 8-channel DWDM link

Given data

Parameter	Value
WDM grid	8 channels, 100 GHz (≈ 0.8 nm) spacing, around 1550 nm
Link	6 amplified spans × 80 km SSMF = 480 km
Data rate	25 Gb/s NRZ per channel, $B \approx 0.7 R_b$
Source data	$E_g = 0.80$ eV, $\tau_r = 10$ ns, $\tau_{nr} = 40$ ns, $f_{3dB} = 20$ GHz
Operating point	$I_{th} = 12$ mA, $\eta_s = 0.3$ mW/mA, $I = 40$ mA
Thermal	$d\lambda/dT = 0.1$ nm/°C, $\Delta T_{ambient} = 40$ °C, tolerance ±0.1 nm
Fiber	$\alpha = 0.2$ dB/km, $\gamma = 1.3$ W^{-1}km^{-1}, $A_{eff} = 80$ μm², $P_{in} = 2$ mW/channel
Scattering	$g_B = 5 \times 10^{-11}$ m/W, $g_R = 6 \times 10^{-14}$ m/W (exponents assumed — see Assumptions)
Soliton	$T_0 = 15$ ps, $\lvert\beta_2\rvert = 20$ ps²/km

Key results at a glance

Quantity	Result
Emission wavelength	1.55 μm (1550 nm)
Internal quantum efficiency	80 %
Output power at 40 mA	8.4 mW = 9.24 dBm
Required NRZ bandwidth / margin	17.5 GHz / 2.5 GHz (14.3 %)
Uncontrolled drift / TEC stability	4.0 nm (5 channel spacings) / ±1 °C (design ±0.1 °C)
L_eff (80 km)	21.17 km
SPM phase: per span / 6 spans	0.055 rad / 0.33 rad
SBS threshold (CW)	1.59 mW — below the 2 mW launch, so SBS must be suppressed
SRS threshold	≈ 1.01 W — 63 × above the 16 mW total
L_D / P_0 for N = 1 / L_NL / N	11.25 km / 68.4 mW / 384.6 km / 0.171

Stage 1 – Interpreting the communication requirement (10 marks)

The link needs 8 independent, grid-locked, low-chirp 25 Gb/s optical carriers that survive 480 km of amplified fiber without nonlinear or dispersive collapse. Working backward from that statement fixes what the transmitter must do.

1.1 Why speed, linewidth and power are all needed

- **High modulation speed.** NRZ at R_b = 25 Gb/s needs B ≈ 0.7 R_b = 17.5 GHz of flat transmitter response. A slower source closes the eye through inter-symbol interference.
- **Narrow linewidth.** Channels sit only 100 GHz (0.8 nm) apart. The source must occupy one slot, with no side modes leaking into neighbours. Narrow optical spectra also reduce dispersion spreading over 480 km (D ≈ 15.7 ps/nm·km × 480 km ≈ 7 530 ps/nm accumulated).
- **High output power.** Each 80 km span loses 16 dB (0.2 dB/km × 80 km). The source must deliver enough power, after modulator and multiplexer losses, to set a 2 mW (3 dBm) per-channel launch that keeps OSNR high through six EDFAs.

1.2 Source characteristics that set wavelength, efficiency and purity

Characteristic	What it controls	Requirement here
Bandgap E_g	Emission wavelength λ ≈ 1.24/E_g	≈ 0.80 eV for 1550 nm (InGaAsP/InP)
Internal quantum efficiency η_i	Fraction of carriers that	High (≥ 70 %)

Characteristic	What it controls	Requirement here
	recombine radiatively	
Threshold current I_th	Onset of stimulated emission, heat, drive power	Low (≈ 10–15 mA)
Slope efficiency η_s	mW of light per mA above threshold	≥ 0.2 mW/mA
Linewidth / SMSR	Spectral purity, crosstalk, dispersion	Single longitudinal mode, SMSR ≥ 30–35 dB, linewidth ≤ 10 MHz

1.3 Modulation, chirp and temperature quantities to control

- Relaxation-oscillation frequency f_R and 3 dB bandwidth (must exceed 17.5 GHz).
- Frequency chirp, set by the linewidth-enhancement factor α_H (3–6 for InP lasers under direct modulation).
- Extinction ratio and eye quality (ER, TDP/TDECQ, eye-mask margin).
- Wavelength drift $d\lambda/dT = 0.1$ nm/°C and the TEC loop that limits it to ±0.1 nm.
- Bias point stability (I = 40 mA, I_th = 12 mA) and RIN.

1.4 Nonlinear quantities that grow with channels, spans and power

- Effective length L_eff and the SPM phase $\varphi_{NL} = \gamma P L_{eff}$, accumulating over 6 spans.
- XPM and FWM, which depend on channel spacing (100 GHz), local dispersion and total power (8 × 2 = 16 mW).
- SBS threshold (per channel, narrow-linewidth sensitive) and SRS threshold (total power, inter-channel tilt).

1.5 Role of the standards

- **ITU-T G.957** defines optical interface parameters at reference point S: spectral width / SMSR, mean launched power range, minimum extinction ratio and the transmitter eye mask. ITU-T G.694.1 (assumed added) fixes the 100 GHz DWDM grid anchored at 193.1 THz.
- **IEEE 802.3bs/bm** define Ethernet PHY transmitter limits for 25.78 GBd lanes: average launch power and OMA per lane, extinction ratio, SMSR ≥ 30 dB, RIN, TDP/TDECQ and eye-mask hit ratio, and centre-wavelength range.
- **ANSI Z136 (Z136.2 for fiber systems)** sets the hazard level of the total launched power, which drives automatic power reduction (APR) and labelling.

1.6 System parameters for soliton/DSP and integration decisions

Soliton feasibility depends on T_0, β_2, γ and P_0 (through L_D, L_NL and N) and on amplifier spacing relative to L_D. DSP compensation (backpropagation) depends on whether the receiver is coherent, on the channel count and on the available processing power. Integration depends on channel count (8), footprint, power per bit and coupling loss.

Stage 1 deliverable – Problem Requirement Statement

> *The transmitter shall generate 8 single-longitudinal-mode carriers on the ITU-T 100 GHz grid near 1550 nm, each held within ±0.1 nm over a 40 °C ambient swing. Each carrier shall be modulated at 25 Gb/s NRZ with ≥ 17.5 GHz bandwidth and minimum chirp. It shall launch 2 mW (3 dBm) per channel (16 mW total) into 6 × 80 km of SSMF, below the effective SBS threshold, with SPM/XPM/FWM penalties held small. Output shall meet ITU-T G.957/G.694.1 and IEEE 802.3bs/bm transmitter limits and ANSI Z136 eye safety, using a compact, low-power, scalable implementation.*

Stage 2 – Optical source investigation and selection (15 marks)

The candidate active material emits at 1550 nm with 80 % internal quantum efficiency, and only a DFB laser turns that material into a single-mode, grid-compatible 25 Gb/s carrier.

2.1 Recombination processes and bandgap

Three recombination paths compete in the active region:

- **Radiative recombination** – an electron in the conduction band recombines with a hole and emits a photon of energy $h\nu \approx E_g$. It occurs spontaneously (LEDs, below threshold) or by stimulation (lasers, above threshold).
- **Non-radiative recombination** – Shockley–Read–Hall (defect/trap) recombination and surface recombination release energy as heat.
- **Auger recombination** – the energy is handed to a third carrier. It is significant in narrow-gap 1550 nm InGaAsP and raises threshold with temperature.

The bandgap fixes the photon energy and so the wavelength:

$$\lambda\,(\mu m) \approx \frac{1.24}{E_g\,(eV)} = \frac{1.24}{0.80} = 1.55\ \mu m = 1550\ nm$$

This falls in the 1550 nm minimum-loss window and inside the EDFA C-band (1530–1565 nm). The material is therefore InGaAsP (or InGaAlAs) lattice-matched to InP.

The internal quantum efficiency is the radiative fraction of all recombination:

$$\eta_i = \frac{1/\tau_r}{1/\tau_r + 1/\tau_{nr}} = \frac{\tau_{nr}}{\tau_r + \tau_{nr}} = \frac{40\ ns}{10\ ns + 40\ ns} = 0.80 = 80\ \%$$

The effective carrier lifetime is $\tau = \tau_r \tau_{nr} / (\tau_r + \tau_{nr}) = (10 \times 40)/50 = 8$ ns. An 80 % η_i is good for InP material and supports the high slope efficiency (0.3 mW/mA) of the candidate laser.

2.2 Comparison of candidate sources

Parameter	LED	FP laser	DBR laser	DFB laser	VCSEL (1550 nm)	QD laser
Output power into SMF	≈ 10–100 μW	1–10 mW	5–20 mW	5–40 mW	0.5–2 mW	1–20 mW
Spectral width	50–100 nm	3–10 nm (multimode, ≈ 1.1 nm mode	< 1 MHz–10 MHz, tunable	1–10 MHz, single mode	10–100 MHz, polarization	Broad gain, often multimode

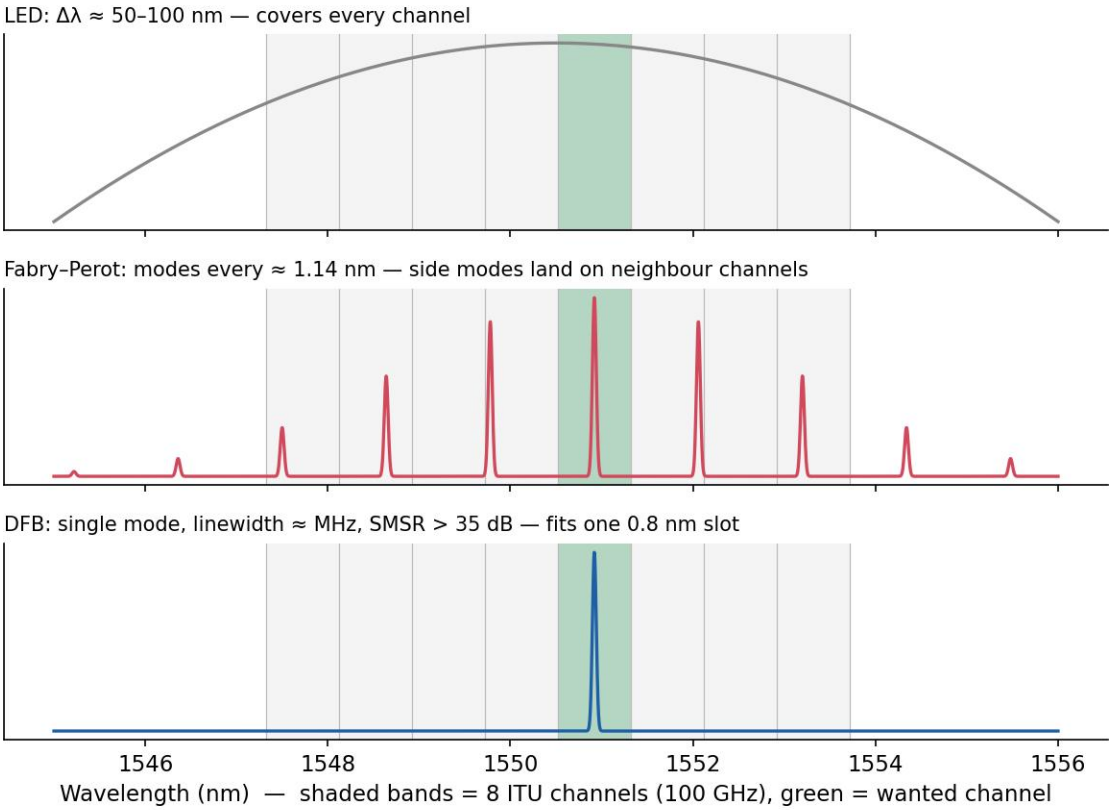
Parameter	LED	FP laser	DBR laser	DFB laser	VCSEL (1550 nm)	QD laser
		spacing)			-unstable	comb
SMSR	n/a	< 10 dB	35–45 dB	35–50 dB	30–40 dB	Low (FP type)
Modulation speed	< 0.5 GHz	5–10 GHz	10–20 GHz	15–30 GHz	10–25 GHz	10–15 GHz
Efficiency / threshold	No threshold, low efficiency	I_th 10–30 mA	I_th 10–30 mA	I_th 8–20 mA	I_th < 2 mA, very efficient	Lowest I_th, temperature -stable
Fiber coupling	Poor (Lambertian)	Good	Good	Good (with isolator)	Excellent (circular beam)	Good
DWDM suitability	No	No	Yes (tunable)	Yes (best fixed-λ)	Short reach only	Mainly 1310 nm, datacom

The Fabry–Perot result is decisive against FP. For a cavity L ≈ 300 μm with group index n ≈ 3.5:

$$\Delta\lambda_{mode} = \frac{\lambda^2}{2nL} = \frac{(1.55\times10^{-6})^2}{2(3.5)(300\times10^{-6})} \approx 1.14\ nm$$

FP side modes therefore land on the neighbouring 0.8 nm-spaced channels, as Fig. 2 shows.

Fig. 2 Source spectra against the 100 GHz DWDM grid



2.3 Threshold condition, linewidth and SMSR

Oscillation starts when round-trip gain equals round-trip loss (amplitude condition) and the round-trip phase is a multiple of 2π (phase condition):

$$\Gamma g_{th} = \alpha_i + \frac{1}{2L}\ln\!\left(\frac{1}{R_1 R_2}\right), \qquad 2\beta L = 2m\pi$$

Worked example (assumed $L = 300$ µm, $R_1 = R_2 = 0.32$, $\alpha_i = 10$ cm^{-1}): mirror loss = $(1/0.06$ cm$)\ln(1/0.1024) = 38$ cm^{-1}, so $\Gamma g_{th} \approx 48$ cm^{-1}. In a DFB, the Bragg grating makes the loss strongly wavelength-selective, so only the mode at $\lambda_B = 2\, n_{eff}\, \Lambda$ reaches threshold.

Effect on an 8-channel DWDM link:

- **Linewidth.** The Schawlow–Townes linewidth is broadened by $(1 + \alpha_H^2)$. A DFB (1–10 MHz) is far narrower than the 25 GHz signal bandwidth, so the modulated spectrum is set by the data, not the laser. However, a narrow linewidth lowers the SBS threshold (Stage 4).
- **SMSR.** Any side mode carries part of the power into another channel slot. SMSR ≥ 35 dB keeps that crosstalk ≤ −35 dB, well below the ≈ −20 dB level that causes a measurable penalty. Mode-partition noise is also suppressed.

2.4 Selection and alternative

The DFB laser is selected on five technical parameters:

1. **Wavelength:** $E_g = 0.80$ eV gives 1550 nm; the grating sets λ_B on an ITU channel.
2. **Spectral purity:** single longitudinal mode, SMSR ≥ 35 dB, linewidth ≈ 1–10 MHz ≪ 0.8 nm slot.

3. **Power:** 8.4 mW at 40 mA (Stage 3) is enough to cover modulator and mux losses.
4. **Speed:** 20 GHz bandwidth exceeds the 17.5 GHz needed for 25 Gb/s NRZ.
5. **Efficiency:** η_i = 80 %, I_{th} = 12 mA, η_s = 0.3 mW/mA give low drive power.

Credible alternative – widely tunable sampled-grating DBR laser. It becomes preferable when the operator wants one universal ("colourless") part that can be set to any of the 8 (or 80) channels, for sparing, reconfigurable ROADM networks or software-defined wavelength assignment. The penalty is more complex control (3–4 tuning currents) and a wavelength locker.

Design Decision 1 – Proposed optical source

> ***I recommend a 1550 nm InGaAsP/InP multiple-quantum-well DFB laser (one per ITU channel, operated CW and externally modulated) optical source because*** *the 0.80 eV bandgap places emission at 1.55 µm in the EDFA C-band, the 80 % internal quantum efficiency and 0.3 mW/mA slope give 8.4 mW at 40 mA, the grating delivers single-mode output with ≥ 35 dB SMSR and MHz linewidth that fits a 0.8 nm slot, and the 20 GHz bandwidth covers the 17.5 GHz NRZ requirement. LED, FP, VCSEL and QD sources each fail at least one of these (spectral width, side modes, power or wavelength).*

Stage 3 – Modulation, chirp and spectral-stability strategy (20 marks)

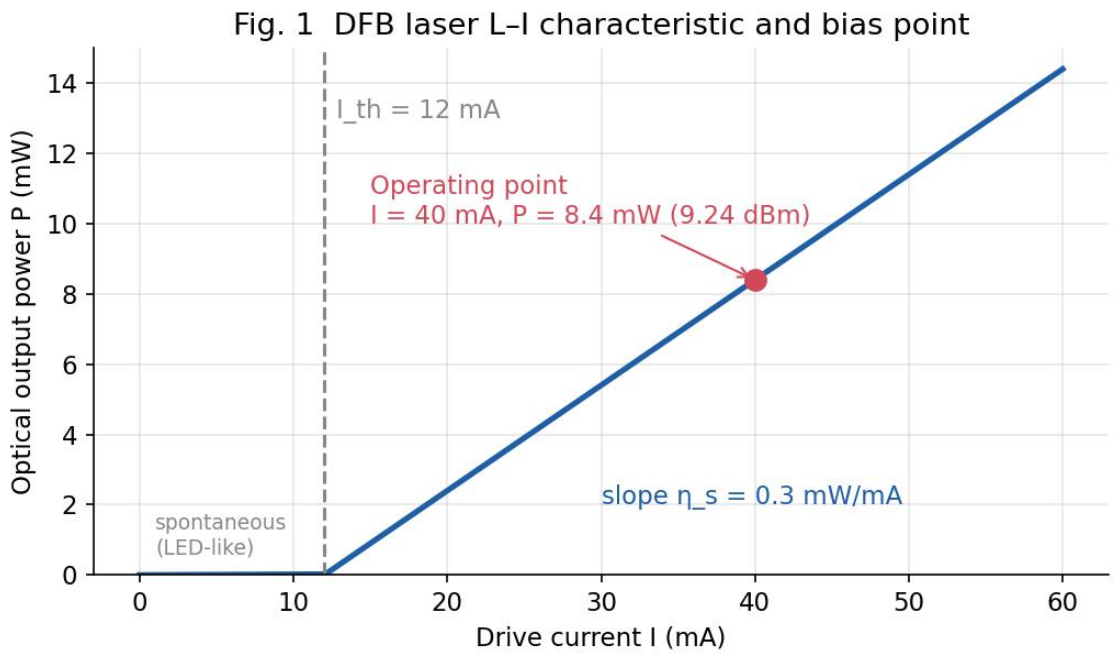
The DFB delivers 8.4 mW (9.24 dBm) with a 2.5 GHz bandwidth margin, but direct-modulation chirp and a 4 nm uncontrolled thermal drift rule out a simple directly modulated, free-running laser.

3.1 Output power at the bias point

$$P = \eta_s (I - I_{th}) = 0.3\ \tfrac{mW}{mA} \times (40 - 12)\ mA = 8.4\ mW$$

$$P_{dBm} = 10\log_{10}\!\left(\frac{8.4\ mW}{1\ mW}\right) = 9.24\ dBm$$

The bias is I/I_{th} = 3.3 × threshold, well clear of the threshold knee, which keeps relaxation-oscillation frequency high and avoids turn-on delay.



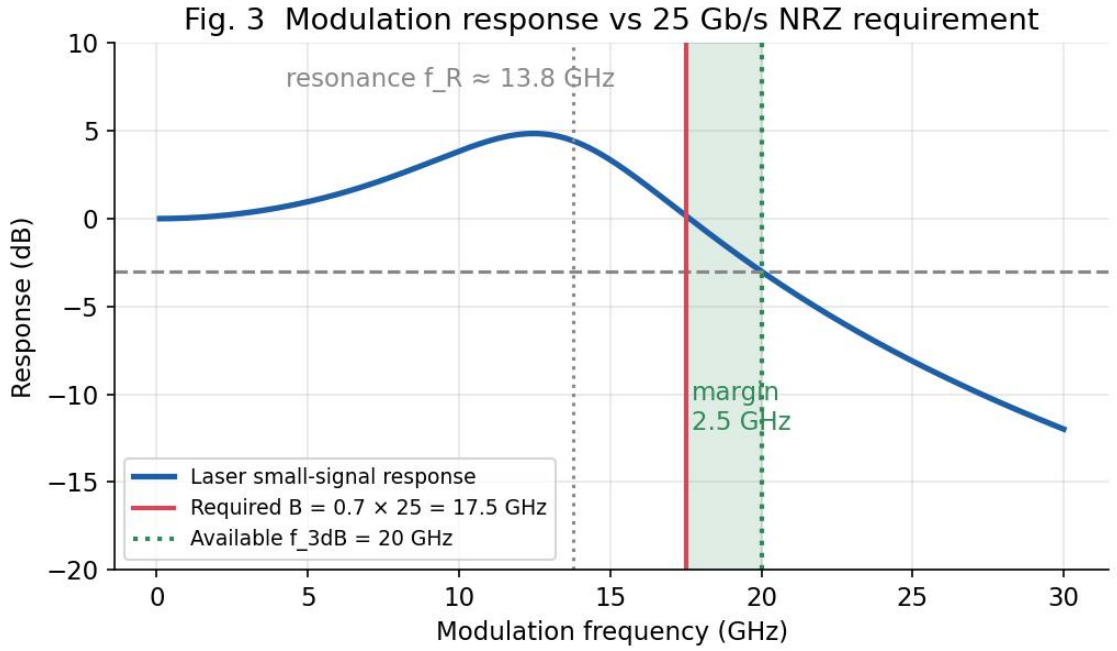
Power budget to the fiber (assumed losses): 9.24 dBm (DFB) − 6.0 dB (MZM: 3 dB insertion + 3 dB NRZ modulation loss at quadrature) − 2.5 dB (AWG multiplexer) − 1.5 dB (chip-to-fiber coupling) = −0.76 dBm per channel. A booster EDFA with ≈ 3.8 dB gain then sets the required launch of 3.01 dBm (2 mW) per channel, with per-channel VOAs for equalisation.

3.2 Bandwidth requirement and margin

$$B \approx 0.7\,R_b = 0.7 \times 25\ \text{Gb/s} = 17.5\ \text{GHz}$$

$$\text{Margin} = f_{3dB} - B = 20 - 17.5 = 2.5\ \text{GHz}\ (14.3\ \%), \quad f_{3dB}/B = 1.14$$

The 20 GHz bandwidth is **sufficient**. With Ethernet 64B/66B line rate (25.78 GBd) the need is 18.05 GHz and the margin 1.95 GHz, still positive. The margin is modest, so the external modulator and driver must also be specified at ≥ 20 GHz.



3.3 Rate equations and relaxation oscillation

Carrier density N and photon density S obey:

$$\frac{dN}{dt} = \frac{I}{qV} - \frac{N}{\tau_e} - v_g\, a\,(N - N_{tr})\, S$$

$$\frac{dS}{dt} = \Gamma v_g\, a\,(N - N_{tr})\, S - \frac{S}{\tau_p} + \Gamma \beta_{sp} \frac{N}{\tau_r}$$

The phase (chirp) equation adds $d\varphi/dt = (\alpha_H/2)[\Gamma v_g a(N - N_{tr}) - 1/\tau_p]$.

Linearising about the bias gives a second-order (resonant) response with relaxation-oscillation frequency:

$$f_R \approx \frac{1}{2\pi}\sqrt{\frac{v_g\, a\, S_0}{\tau_p}} \;\propto\; \sqrt{I - I_{th}}, \qquad f_{3dB} \approx 1.55\, f_R$$

With f_{3dB} = 20 GHz, $f_R \approx 20/1.55 \approx 12.9$ GHz (weak damping); with moderate damping (Fig. 3) $f_R \approx 13.8$ GHz.

Aspect	Small-signal modulation	Large-signal modulation
Drive	Small sinusoid around a fixed bias above threshold	Full on–off swing (NRZ data)
Model	Linearised rate equations, transfer function H(f)	Full nonlinear rate equations
f_R behaviour	Fixed, sets resonance peak and f_{3dB}	Varies through the bit as S changes; f_R falls at the '0' level
Effects	Resonance peak, roll-off above f_R	Turn-on delay, ringing/overshoot, pattern effects, strong transient chirp

If the laser were switched from below threshold, the turn-on delay would be $t_d = \tau_e \ln[I/(I - I_{th})] = 8$ ns × ln(40/28) = **2.85 ns**, which is 71 bit periods at 25 Gb/s ($T_b = 40$ ps). The laser must therefore always be biased above threshold.

3.4 Direct vs external modulation

Direct modulation produces chirp:

$$\Delta\nu(t) = \frac{\alpha_H}{4\pi}\left[\frac{d}{dt}\ln P(t) + \kappa P(t)\right]$$

Estimate (assumed $\alpha_H = 4$, 6 dB extinction, 20 ps edges): transient chirp ≈ (4/4π) × ln(4)/20 ps ≈ 22 GHz ≈ 0.18 nm. The dispersion limit for a chirped source is $L < 1/(4\,B\,D\,\Delta\lambda) = 10$ ps/(15.7 ps/nm·km × 0.18 nm) ≈ **3.5 km**, against a 480 km requirement.

Criterion	Direct modulation (DML)	External modulation (CW DFB + MZM/EAM)
Chirp	High ($\alpha_H \approx 3$–6), ≈ 0.1–0.2 nm transient	Near zero (push-pull MZM); small for EAM ($\alpha \approx 0$–0.7)
Bandwidth	Limited by f_R (20 GHz here, 2.5 GHz margin)	Set by modulator (25–40 GHz available)
Dispersion interaction	Chirp × D broadens pulses; SPM–chirp interplay	Spectrum ≈ data bandwidth only; predictable, compensable
Reach at 25 Gb/s on SSMF	≈ 3–5 km uncompensated	≈ 5–6 km uncompensated (±100 ps/nm), hundreds of km with DCMs
Cost / power	Lowest	Extra modulator, driver and ≈ 6 dB loss

Only external modulation produces a spectrum clean enough to be dispersion-managed over 480 km.

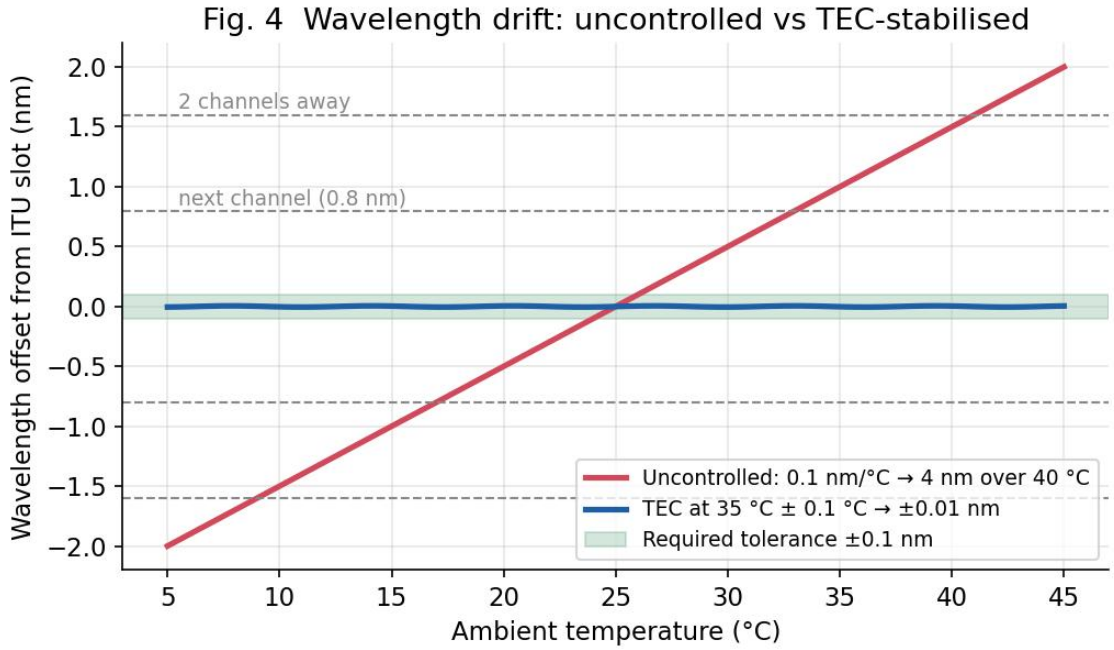
3.5 Temperature drift and wavelength control

$$\Delta\lambda = \frac{d\lambda}{dT}\,\Delta T = 0.1\ \tfrac{nm}{^\circ C} \times 40\ ^\circ C = 4.0\ nm$$

This is 4.0/0.8 = **5 channel spacings**, so a free-running laser would sweep across five neighbouring channels. The TEC must hold:

$$\Delta T_{max} = \frac{\pm 0.1\ nm}{0.1\ nm/^\circ C} = \pm 1\ ^\circ C$$

The TEC loop must reject a 40 °C swing down to ±1 °C, a thermal rejection of at least 20 (26 dB). The design target is ±0.1 °C (±0.01 nm), giving 10× margin for ageing and locker error.



Criterion	TEC (temperature) control	Injection-current control
Tuning rate	0.1 nm/°C (≈ 12.5 GHz/°C)	≈ 0.01 nm/mA (assumed)
Range	≈ 4–5 nm over 40–50 °C	≈ 0.2–0.3 nm before power/reliability limits
Side effects	Slow (seconds), TEC power ≈ 0.5–2 W	Changes output power (0.3 mW/mA) and chirp
Use	Coarse setting and long-term stability	Fine trim only

Correcting 4 nm by current would need ≈ 400 mA, which is impossible. TEC control, closed by an etalon wavelength locker, is the only workable primary method.

Design Decision 2 – Modulation configuration

> ***I recommend external (zero-chirp push-pull Mach–Zehnder) modulation and TEC wavelength control (TEC at 35 °C ± 0.1 °C, closed by an etalon wavelength locker, with micro-heater/current fine trim) because*** *the CW DFB supplies 8.4 mW (9.24 dBm), enough to absorb the ≈ 10 dB modulator/mux/coupling loss before a booster EDFA; the 17.5 GHz NRZ bandwidth is covered with a 2.5 GHz margin by both laser and modulator; direct-modulation chirp (≈ 0.18 nm) would limit reach to ≈ 3.5 km; and the 4 nm uncontrolled drift (5 channel spacings) must be cut to within ±0.1 nm, which only a TEC holding ±1 °C (design ±0.1 °C) can do.*

Stage 4 – Investigating and controlling optical nonlinearities (20 marks)

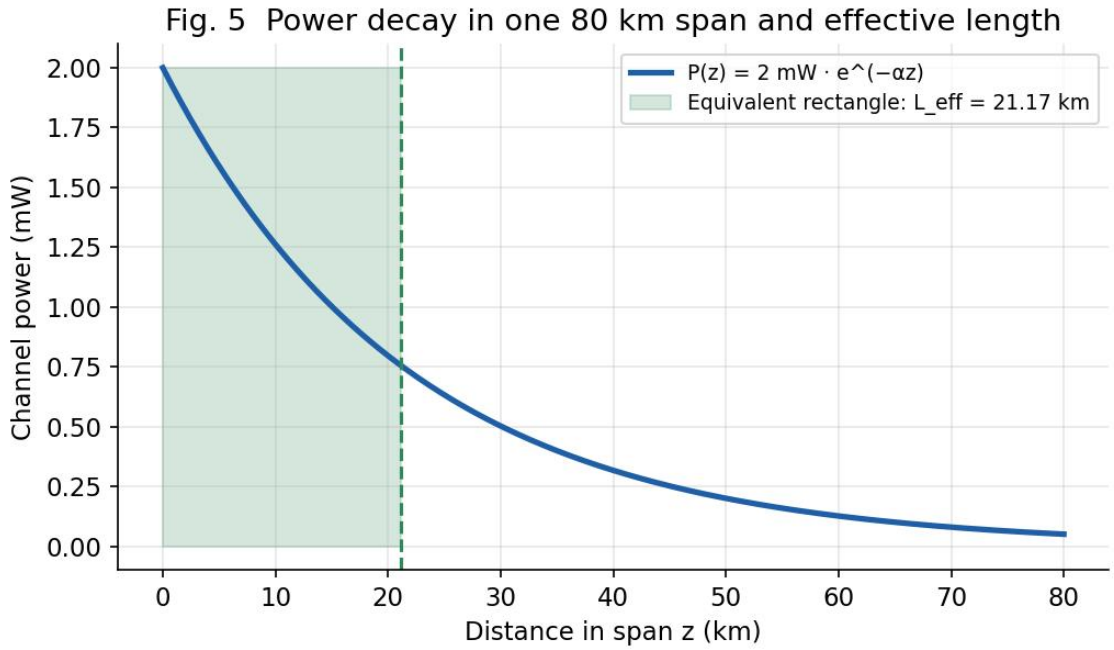
At 2 mW per channel the Kerr effects are small (0.33 rad SPM over 480 km), SRS is negligible, but the CW SBS threshold (1.59 mW) is below the launch power, so SBS is the one mechanism that must be actively suppressed.

4.1 Effective length and SPM phase

Convert attenuation to linear units: $\alpha = 0.2$ dB/km $\div$ 4.343 = 0.04605 km^{-1}, so $\alpha L = 0.04605 \times 80 = 3.684$.

$$L_{eff} = \frac{1 - e^{-\alpha L}}{\alpha} = \frac{1 - e^{-3.684}}{0.04605\ km^{-1}} = \frac{1 - 0.02512}{0.04605} = 21.17\ km$$

Almost all nonlinear interaction happens in the first ≈ 21 km after each amplifier (Fig. 5).



$$\phi_{NL,span} = \gamma P L_{eff} = 1.3\ W^{-1}km^{-1} \times 0.002\ W \times 21.17\ km = 0.0550\ rad$$

$$\phi_{NL,total} = N_s\, \gamma P L_{eff} = 6 \times 0.0550 = 0.330\ rad$$

The total SPM phase is well below the commonly used ≈ 1 rad design ceiling, so SPM alone causes only a small penalty (well under 1 dB) for NRZ.

4.2 XPM and FWM with 100 GHz spacing

XPM. The phase on channel i depends on all others: $\varphi_i = \gamma L_{eff} (P_i + 2 \Sigma_{j \neq i} P_j)$. Worst case (no walk-off) for 7 neighbours at 2 mW: $2 \times 1.3 \times 0.014 \times 21.17 = 0.77$ rad per span. In practice dispersion makes channels walk off. With $D \approx 15.7$ ps/(nm·km) (from $|\beta_2| = 20$ ps²/km) and $\Delta\lambda = 0.8$ nm, adjacent channels separate by 12.6 ps/km, so a 40 ps bit walks past its neighbour in $L_W \approx 40/12.6 \approx 3.2$ km $\ll L_{eff}$. XPM is averaged out, leaving a modest, mostly timing/amplitude penalty. XPM grows with total power and shrinks as spacing and local dispersion grow.

FWM. Three channels at f_i, f_j, f_k create a product at f_i + f_j − f_k. On an equally spaced 100 GHz grid these products fall exactly on other channels, causing coherent in-band crosstalk that no filter can remove. FWM efficiency depends on phase mismatch:

$$\Delta\beta \approx |\beta_2|(2\pi\Delta f)^2 = 20\times10^{-24}\ \tfrac{s^2}{km}\,(2\pi\times10^{11})^2 = 7.90\ km^{-1}$$

$$\eta_{FWM} \approx \frac{\alpha^2}{\alpha^2 + \Delta\beta^2} = \frac{0.04605^2}{0.04605^2 + 7.90^2} = 3.4\times10^{-5}\;(-44.7\ dB)$$

On SSMF the high local dispersion keeps FWM ≈ 45 dB down. FWM would become severe on dispersion-shifted fiber ($\Delta\beta \to 0$) or if the channel spacing were reduced. FWM power scales as P^3, so it rises 9 dB for every 3 dB increase in launch power.

4.3 SBS and SRS thresholds

With $A_{eff} = 80 \times 10^{-12}$ m², $L_{eff} = 21\,170$ m:

$$P_{th}^{SBS} \approx \frac{21\,A_{eff}}{g_B\,L_{eff}} = \frac{21 \times 80\times10^{-12}}{5\times10^{-11} \times 21\,170} = \frac{1.68\times10^{-9}}{1.0585\times10^{-6}} = 1.59\ mW\ (2.0\ dBm)$$

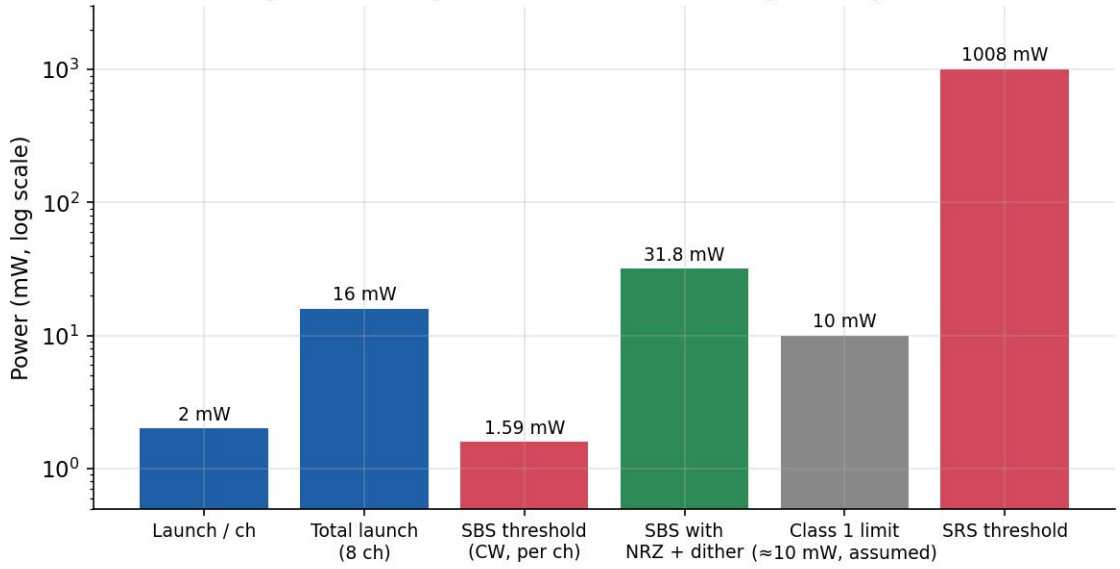
$$P_{th}^{SRS} \approx \frac{16\,A_{eff}}{g_R\,L_{eff}} = \frac{16 \times 80\times10^{-12}}{6\times10^{-14} \times 21\,170} = \frac{1.28\times10^{-9}}{1.270\times10^{-9}} = 1.01\ W\ (30.0\ dBm)$$

Check	Power	Threshold	Ratio	Verdict
SBS, per channel (CW linewidth)	2 mW	1.59 mW	1.26 × above	**Exceeded – must suppress**
SBS, with NRZ (×2) and phase dither (×10)	2 mW	≈ 32 mW	16 × below	Safe after mitigation
SRS, total 8-channel power	16 mW	1.01 W	63 × below (18 dB)	Safe

SBS is checked per channel because the Brillouin gain bandwidth (≈ 20 MHz) is far narrower than the 100 GHz spacing, so each channel scatters independently. SRS is checked on total power because its ≈ 13 THz gain bandwidth couples all channels.

SRS tilt estimate. The 8 channels span only 700 GHz, where Raman gain is ≈ 5 % of its peak. Summing over 7 pump channels (Δf = 100…700 GHz, total 2.8 THz) gives a depletion of the shortest-wavelength channel of about 0.34 % per span, ≈ 2 % (≈ 0.09 dB) over 6 spans. This is negligible.

Fig. 6 Launch power vs SBS, SRS and eye-safety limits



4.4 SBS vs SRS

Property	SBS (Brillouin)	SRS (Raman)
Physical origin	Light scatters off an acoustic phonon (electrostriction)	Light scatters off an optical phonon (molecular vibration)
Frequency shift	≈ 11 GHz (≈ 0.09 nm)	≈ 13 THz (≈ 100 nm)
Gain bandwidth	≈ 20–50 MHz	≈ 10–13 THz (broad)
Peak gain	≈ 5×10^{-11} m/W	≈ 6×10^{-14} m/W (≈ 1000× weaker)
Direction	Backward only	Forward and backward
Power transfer	Signal → backward Stokes wave; transmitted power saturates, back-reflection and noise rise	Short-λ channels → long-λ channels (spectral tilt), plus Raman crosstalk
DWDM effect	Per-channel power clamp, noise; independent per channel	Inter-channel tilt and crosstalk; depends on total power and band width
Threshold here	1.59 mW per channel	1.01 W total

Linewidth and SBS. SBS gain is reduced when the source spectrum $\Delta\nu_s$ exceeds the Brillouin bandwidth $\Delta\nu_B$:

$$P_{th}^{SBS}(\Delta\nu_s) \approx P_{th}^{SBS}(0)\left(1 + \frac{\Delta\nu_s}{\Delta\nu_B}\right)$$

A 1–10 MHz DFB is narrower than $\Delta\nu_B \approx 20$ MHz, so it sees nearly the full Brillouin gain. Two effects raise the threshold. NRZ data leaves only about half the power in the carrier (≈ ×2). Phase dithering of the

DFB (a few tones at 100 kHz–2 GHz, ≈ 200 MHz spread, assumed) spreads it over ≈ 10 Brillouin bandwidths (≈ ×10). Together they lift the effective threshold to ≈ 32 mW.

4.5 Eye safety and mitigation

Total launch = 8 × 2 mW = 16 mW = **12.04 dBm**.

Assumed limits (approximate, for 1550 nm under ANSI Z136.2 / IEC 60825-2 hazard levels): Hazard Level 1 ≈ 10 mW (10 dBm); Hazard Level 1M ≈ 136 mW (21.3 dBm).

The 16 mW total **exceeds Hazard Level 1 by 2 dB** but sits **9.3 dB below the Hazard Level 1M limit**. The link is acceptable in restricted locations (central offices, cable vaults) provided:

- automatic power reduction (APR/ALS, ITU-T G.664) shuts down or reduces the booster within ≈ 1–3 s of a loss-of-signal at the far end (fiber break or open connector);
- connectors are labelled for Hazard Level 1M, and viewing with optical aids is prohibited.

Two practical methods to reduce nonlinear penalties **without lowering received power**:

1. **SBS phase dithering (linewidth broadening) of each DFB.** It raises the SBS threshold ≈ 10× while keeping the 2 mW launch. Phase (not amplitude) dither adds no intensity noise, and the residual dispersion-converted AM is small.
2. **Dispersion management that keeps local dispersion high.** Use SSMF (D ≈ 16 ps/nm·km) in every span, and place the dispersion-compensating modules in the EDFA mid-stage with slight per-span under-compensation (≈ +20 ps/nm residual per span). High local dispersion keeps FWM at ≈ −45 dB and walks XPM off in ≈ 3 km, while the end-to-end residual is trimmed near zero at the receiver.

Supporting measures: per-channel power pre-emphasis (VOAs) to equalise tilt, and hybrid Raman/EDFA amplification so OSNR stays high without raising launch power.

Design Decision 3 – Nonlinearity-control strategy

> ***I recommend a fixed 2 mW (3 dBm) per-channel launch with SBS phase-dither linewidth broadening, high-local-dispersion SSMF with mid-stage DCM per span (slight under-compensation), per-channel power equalisation, and APR for Hazard Level 1M eye safety as the nonlinear-control strategy because*** *the calculated SPM phase is only 0.055 rad per span and 0.33 rad over 6 spans, FWM efficiency is ≈ −45 dB and XPM walk-off length (≈ 3.2 km) is much shorter than L_eff (21.17 km); SRS (1.01 W threshold vs 16 mW total) is negligible; but the 1.59 mW CW SBS threshold sits below the 2 mW launch and must be raised to ≈ 32 mW by dithering; and the 16 mW total exceeds the ≈ 10 mW Class/Hazard Level 1 limit, so APR is required.*

Stage 5 – Soliton transmission, digital backpropagation and photonic integration (20 marks)

Soliton transmission is not feasible (N = 0.171 at 2 mW; N = 1 would need 68.4 mW per channel), digital backpropagation is not applicable to a 25 Gb/s NRZ direct-detection link, and a monolithic InP transmitter PIC is the best implementation.

5.1 Physical basis of a soliton

In the anomalous-dispersion regime ($\beta_2 < 0$, $D > 0$, true for SSMF at 1550 nm), dispersion makes higher frequencies travel faster, producing a negative chirp across the pulse. Kerr SPM ($n = n_0 + n_2 I$) produces a positive chirp. When the two chirps cancel exactly, the pulse keeps its sech² shape. This balance is described by the nonlinear Schrödinger equation:

$$\frac{\partial A}{\partial z} + \frac{i\beta_2}{2}\frac{\partial^2 A}{\partial T^2} = i\gamma |A|^2 A, \qquad A(0,T) = \sqrt{P_0}\,\mathrm{sech}(T/T_0)$$

5.2 Dispersion length and fundamental-soliton power

$$L_D = \frac{T_0^2}{|\beta_2|} = \frac{(15\ ps)^2}{20\ ps^2/km} = \frac{225}{20} = 11.25\ km$$

For N = 1, $L_{NL} = L_D$, so:

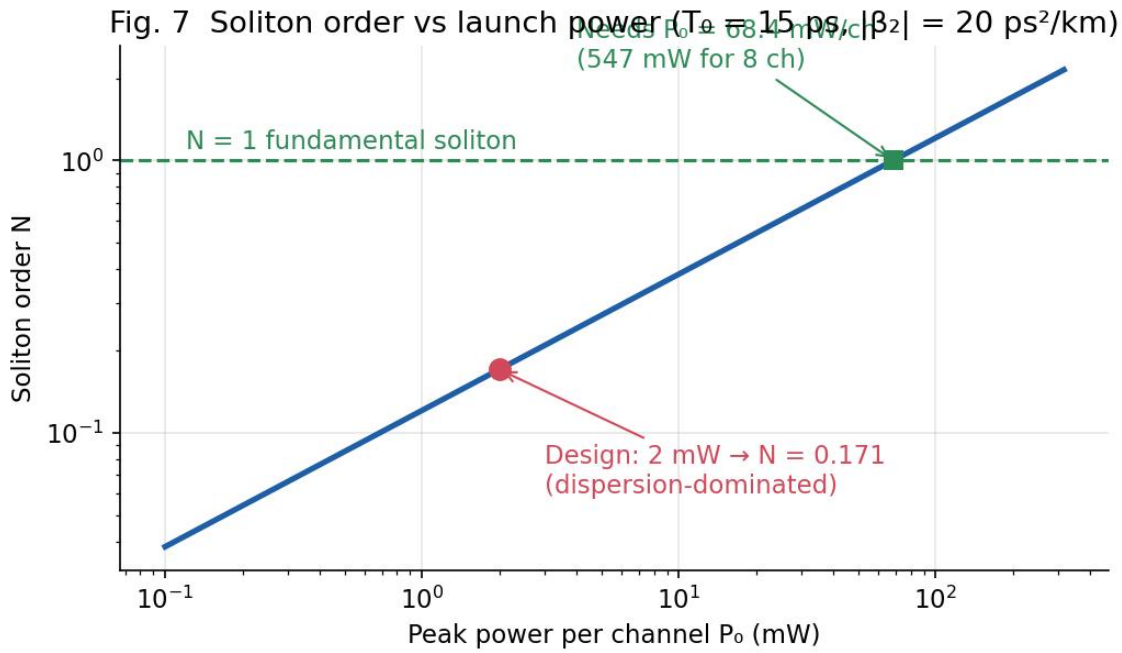
$$P_0 = \frac{1}{\gamma L_D} = \frac{1}{1.3\ W^{-1}km^{-1} \times 11.25\ km} = 0.0684\ W = 68.4\ mW\ (18.35\ dBm)$$

5.3 Soliton order at the design launch power

$$L_{NL} = \frac{1}{\gamma P_0} = \frac{1}{1.3 \times 0.002} = 384.6\ km$$

$$N = \sqrt{\frac{L_D}{L_{NL}}} = \sqrt{\frac{11.25}{384.6}} = \sqrt{0.02925} = 0.171$$

Fig. 7 Soliton order vs launch power (T_0 = 15 ps, $|\beta_2|$ = 20 ps²/km)



Decision: soliton transmission is not feasible.

- N = 0.171 ≪ 1: the link is dispersion-dominated (L_{NL} is 34 × L_D), so pulses spread linearly; there is not enough nonlinearity to form a soliton.
- Forcing N = 1 needs 68.4 mW per channel (547 mW for 8 channels). That is 43 × the CW SBS threshold, 17 × the dithered SBS threshold, 4 × the Hazard Level 1M limit, and would make FWM ($\propto P^3$) severe.

- Amplifier spacing L_A = 80 km is 7.1 × L_D. Guiding-centre (path-averaged) solitons need L_A ≪ L_D, so periodic loss/gain would destabilise them and shed dispersive waves.
- Solitons are an RZ format. A 15 ps T_0 gives a 26.4 ps FWHM in a 40 ps slot (q_0 = T_B/2T_0 ≈ 1.3), so adjacent solitons would interact strongly; $q_0 \geq 4$–5 is normally required. The specified format is NRZ.

5.4 Digital backpropagation (DBP)

DBP solves the NLSE in reverse in the receiver DSP (split-step Fourier: steps with $-\beta_2$ and $-\gamma$). It undoes the deterministic interaction of dispersion and Kerr nonlinearity, so it compensates CD, SPM, and (if all channels are backpropagated together) XPM and FWM.

Practical limitations:

- It needs the full optical field (amplitude and phase), which requires a coherent receiver. The 25 Gb/s NRZ direct-detection receivers here measure only intensity, so DBP is impossible.
- Full-field DBP of 8 channels requires sampling ≈ 800 GHz of optical bandwidth with many steps per span. The computation and power are very high. Single-channel DBP cannot remove XPM/FWM from neighbours.
- It cannot undo stochastic effects: ASE–signal nonlinear noise and PMD.

Decision: DBP is not appropriate for this 8-channel link. With φ_{NL} = 0.33 rad there is little nonlinear distortion to recover. Optical dispersion management plus a receiver linear equaliser (FFE/DFE or tunable DCM) is sufficient. DBP would become relevant only if the link were upgraded to coherent DP-QPSK/16QAM.

5.5 Standards and how the transmitter is checked

Standard	Transmitter parameters specified	How the design is checked
ITU-T G.957 (SDH interfaces, at reference point S)	Source type (SLM), max −20 dB spectral width, min SMSR (≥ 30 dB), mean launched power range, min extinction ratio, eye-diagram mask	OSA: SMSR ≥ 35 dB, −20 dB width; power meter: launch power; DCA with mask at reference receiver
ITU-T G.694.1 (assumed added)	100 GHz grid anchored at 193.1 THz, frequency accuracy	Wavelength meter across 5–45 °C: within ±0.1 nm (±12.5 GHz)
IEEE 802.3bm / 802.3ba-type 25.78 GBd lanes	Avg launch power and OMA per lane, ER, SMSR ≥ 30 dB, RIN (e.g. ≤ −130 dB/Hz), TDP, eye-mask hit ratio, optical return-loss tolerance	DCA with Bessel-Thomson reference receiver; RIN analyser; TDP via BER-vs-power with reference link
IEEE 802.3bs (200G/400G)	Same set for PAM4 lanes: OMA_outer, TDECQ, ER, SMSR, RIN, transmitter reflectance	Same bench with TDECQ software; relevant if lanes are upgraded to 26.56/53.13 GBd PAM4

Standard	Transmitter parameters specified	How the design is checked
ANSI Z136.2 / IEC 60825-2, ITU-T G.664	Hazard level of accessible power, APR timing	Measure total power at connector; test APR response on fiber-break simulation

Note: G.957 was written for SDH up to 2.5 Gb/s, so its parameter definitions (not its numeric limits) are applied here; the 25 Gb/s numeric limits come from the IEEE 802.3 lane specifications.

5.6 Discrete vs InP vs Si-photonics implementation

Criterion	Discrete (8 × TOSA: DFB + $LiNbO_3$/InP MZM, external AWG)	Monolithic InP PIC (8 DFB + 8 MZM/EAM + AWG on one chip)	Si photonics (hybrid/heterogeneous InP laser + Si MZM + Si AWG)
Footprint	Largest (8 packages + mux, rack-unit scale)	≈ 10× smaller, single chip/package	Smallest, CMOS-dense
Power consumption	Highest: 8 TECs (≈ 1–2 W each), 8 drivers	One shared TEC + micro-heaters; lower	Lowest modulator drive, but Si MZM needs higher V_π and laser heat must be managed
Optical coupling	16+ fiber couplings, highest loss and alignment cost	One fiber coupling; on-chip waveguides	Laser-to-Si coupling (hybrid/bonded) and edge/grating couplers add 1–3 dB
Reliability	Mature, individually replaceable lasers	Mature (field-proven DWDM PICs); one laser failure can scrap the chip unless spares are included	Laser bonding and thermal cross-talk still maturing
Scalability	Linear cost per channel	Excellent to 10–40 channels	Excellent; wafer-scale, co-packaged optics
Performance	Best individual specs ($LiNbO_3$ zero chirp)	Native lasers, good modulators, thermal cross-talk managed	Weaker high-power/linewidth; Si MZM loss higher

How integration affects system performance. Fewer fiber interfaces cut insertion loss and back-reflection (better RIN and linewidth stability). A shared substrate and single TEC tie all 8 wavelengths together, so drift is common-mode and per-channel micro-heaters can trim each laser to the grid. Integration does add thermal and electrical crosstalk between adjacent lasers and modulators, which must be designed out (thermal isolation trenches, driver layout). Lower power per channel also reduces heat, easing TEC load and improving reliability.

Design Decision 4 – Integrated transmitter implementation

> ***I recommend a monolithic InP photonic-integrated transmitter (8 DFB lasers + 8 push-pull InP MZMs + monitor photodiodes + an 8-channel AWG on one chip, on a single TEC with per-laser micro-heaters and an etalon wavelength locker) with optical (DCM) dispersion management and receiver-side linear equalisation, not soliton transmission or DBP, transmitter/integration strategy because*** *InP natively provides the 1550 nm lasers selected in Stage 2 and the external modulators of Stage 3 on one die; it removes 15 of 16 fiber couplings, uses one TEC to hold all channels within ±0.1 nm, scales cleanly to more channels and supports the G.957/IEEE 802.3 transmitter parameters; the low nonlinearity (φ_NL = 0.33 rad, N = 0.171) means no nonlinear DSP is needed, and the IM/DD receivers could not run DBP anyway.*

Stage 6 – Verification and critical evaluation (5 marks)

Every requirement is met; two are met only because of a deliberate mitigation (SBS dithering and APR), and one design choice (no soliton/DBP) is a justified rejection rather than a feature.

Requirement	Specified / target	Achieved	Satisfactory?
Optical source selection	1550 nm, single mode, fits 0.8 nm slot, η_i high	InGaAsP/InP MQW DFB: λ = 1.55 µm, η_i = 80 %, SMSR ≥ 35 dB, 1–10 MHz linewidth	Yes
Modulation bandwidth and output power	≥ 17.5 GHz; enough power for 2 mW launch	20 GHz (margin 2.5 GHz, 14.3 %); 8.4 mW = 9.24 dBm → 2 mW after MZM/AWG/booster	Yes (margin modest; modulator spec ≥ 20 GHz)
Spectral/chirp and wavelength stability	Low chirp; ±0.1 nm over 40 °C	Zero-chirp push-pull MZM (DML would give ≈ 0.18 nm chirp, 3.5 km reach); drift 4.0 nm → ±0.01 nm with TEC ±0.1 °C (need ±1 °C)	Yes
Nonlinear-impairment control and eye safety	φ_NL < 1 rad; below SBS/SRS thresholds; Z136 compliant	φ_NL = 0.33 rad; FWM ≈ −45 dB; SRS 1.01 W ≫ 16 mW; SBS 1.59 mW < 2 mW → raised to ≈ 32 mW by dither; 16 mW > HL1 (10 mW) but < HL1M (136 mW) with APR	Yes, conditional on dither and APR

Requirement	Specified / target	Achieved	Satisfactory?
Soliton / dispersion condition	Decide feasibility; control 7 530 ps/nm accumulated CD	L_D = 11.25 km, $P_0(N=1)$ = 68.4 mW, L_{NL} = 384.6 km, N = 0.171 → not feasible; mid-stage DCMs + receiver trim	Yes (soliton correctly rejected)
Integration / DSP / standards strategy	Compact, scalable, G.957 / IEEE 802.3bs/bm compliant	Monolithic InP PIC, single TEC + locker; linear EQ, no DBP (IM/DD); tests defined for SMSR, ER, OMA, RIN, TDP/TDECQ, mask	Yes

Consistency with the original requirement. The calculated results fit together. The DFB power (8.4 mW) is what makes external modulation affordable. External modulation is what makes dispersion management over 480 km workable. The low launch power that keeps φ_{NL} at 0.33 rad is also what makes solitons impossible (N = 0.171), so the design deliberately operates in the quasi-linear regime.

Critical points and risks.

- The 2.5 GHz bandwidth margin is only 14 %; driver, package and modulator roll-off must be verified together.
- SBS is the binding constraint. If dithering is omitted, the 2 mW launch is 26 % above the 1.59 mW threshold and back-scatter would rise sharply.
- The scattering-gain exponents and eye-safety limits were assumed (see Assumptions); changed values would shift the margins but not the decisions.

Final transmitter design – integrated engineering solution (10 marks)

The adopted design is an 8-channel InP PIC transmitter: CW DFB lasers biased at 40 mA (8.4 mW), zero-chirp MZMs at 25 Gb/s, an on-chip AWG, one TEC at 35 °C ± 0.1 °C with a wavelength locker, SBS dithering, and a booster EDFA with APR that launches 2 mW per channel into a dispersion-managed 6 × 80 km link.

Final parameter / design table

Parameter	Adopted value	Basis
Source	InGaAsP/InP MQW DFB, 8 per PIC	Stage 2
Bandgap / wavelength	0.80 eV → 1.55 µm; channels 193.70–193.00 THz (1547.72–1553.33 nm)	$\lambda = 1.24/E_g$; G.694.1 grid
Internal quantum efficiency	80 % (τ_{eff} = 8 ns)	$\tau_{nr}/(\tau_r + \tau_{nr})$

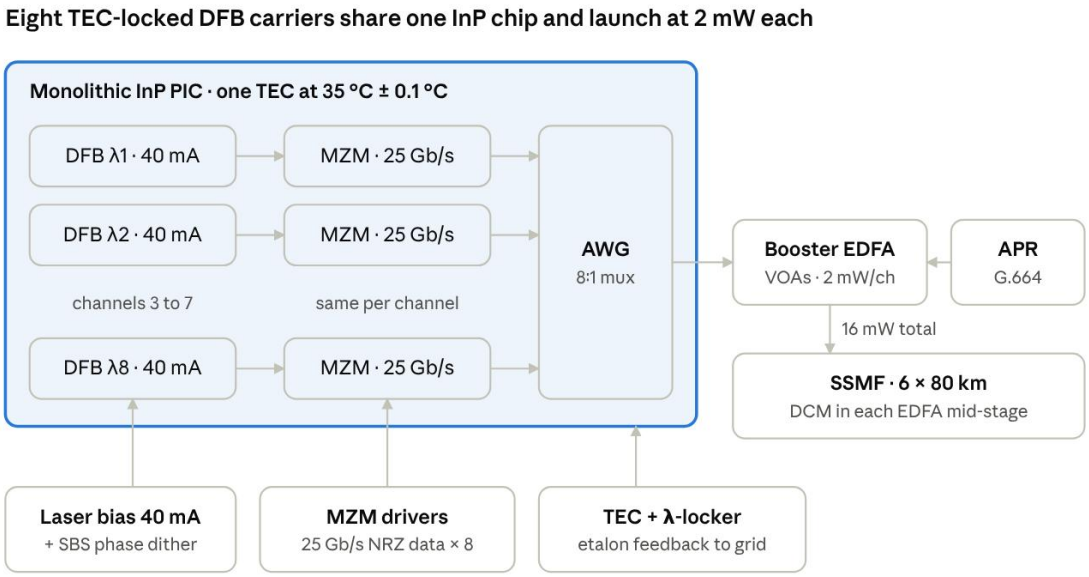
Parameter	Adopted value	Basis
Bias / threshold	40 mA / 12 mA (3.3 × I_th)	Avoids turn-on delay (2.85 ns if switched from 0)
Laser CW output	8.4 mW (9.24 dBm)	η_s(I − I_th)
Modulation	External push-pull InP MZM, 25 Gb/s NRZ, ≥ 20 GHz, ER ≥ 8 dB	Stage 3
Bandwidth need / margin	17.5 GHz / 2.5 GHz (14.3 %)	0.7 R_b
Wavelength control	Single TEC 35 °C ± 0.1 °C (need ±1 °C), etalon locker, micro-heater trim	Drift 4 nm uncontrolled → ±0.01 nm
Launch power	2 mW (3 dBm) per channel; 16 mW (12.04 dBm) total	Booster EDFA + VOAs
SBS control	Phase dither ≈ 200 MHz spread; effective threshold ≈ 32 mW	CW threshold 1.59 mW
Nonlinear phase	0.055 rad/span; 0.33 rad total	γ P L_eff, L_eff = 21.17 km
Dispersion management	Mid-stage DCM per span (≈ +20 ps/nm residual/span), receiver trim	7 530 ps/nm uncompensated
Eye safety	Hazard Level 1M with APR (ITU-T G.664)	16 mW > 10 mW HL1, < 136 mW HL1M
Soliton parameters	L_D = 11.25 km, P_0(N=1) = 68.4 mW, L_NL = 384.6 km, N = 0.171	Not feasible
DSP	Linear FFE/DFE at receiver; no DBP	IM/DD receiver

Channel plan (100 GHz grid):

Ch	Frequency (THz)	Wavelength (nm)
1	193.70	1547.72
2	193.60	1548.51
3	193.50	1549.32
4	193.40	1550.12
5	193.30	1550.92
6	193.20	1551.72

Ch	Frequency (THz)	Wavelength (nm)
7	193.10	1552.52
8	193.00	1553.33

Block diagram of the proposed DWDM transmitter



Proposed 8-channel DWDM transmitter · InP PIC, booster, APR and control loops

Each CW DFB (λ1–λ8, 1547.72–1553.33 nm) feeds its own zero-chirp MZM; the AWG combines all 8 onto one fiber output. Control loops below the chip set the bias and SBS dither, drive the data, and lock every wavelength through one TEC and an etalon locker; the booster EDFA sets 2 mW per channel under APR before the 6-span link.

How the design choices work together

- **Modulation and wavelength control.** The CW DFB runs at a fixed 40 mA, so its wavelength and power never move with the data. The MZM supplies the 25 Gb/s modulation with ≥ 20 GHz bandwidth (2.5 GHz margin over 17.5 GHz) and no chirp. The TEC removes the 4 nm (5-channel) drift down to ±0.01 nm, inside the ±0.1 nm tolerance.
- **Nonlinear control, eye safety and compensation.** Launch is fixed at 2 mW per channel, where SPM totals 0.33 rad, FWM is ≈ −45 dB, and SRS (1.01 W threshold) is irrelevant. SBS, whose 1.59 mW threshold is the binding limit, is lifted to ≈ 32 mW by phase dither. DCMs in the EDFA mid-stage handle dispersion while preserving high local dispersion. APR keeps the 16 mW total safe under Hazard Level 1M.
- **Integration and standards.** One InP chip carries all lasers, modulators, monitor photodiodes and the AWG, with one fiber output and one TEC. Each channel is verified against G.957 parameter definitions (SMSR, spectral width, launch power, ER, eye mask), G.694.1 grid accuracy, and IEEE 802.3bs/bm lane limits (OMA, RIN, TDP/TDECQ, mask hit ratio).

- **Soliton/DSP.** N = 0.171 and L_A/L_D = 7.1 confirm a quasi-linear, dispersion-dominated link. Solitons would need 68.4 mW per channel and DBP needs a coherent receiver, so neither is used.

Final engineering recommendation

Deploy an 8-channel monolithic InP transmitter PIC with TEC-locked 1550 nm DFB lasers biased at 40 mA, external zero-chirp MZMs at 25 Gb/s NRZ, an on-chip AWG and a booster EDFA with APR. Launch 2 mW per channel with SBS phase dithering into SSMF with per-span mid-stage dispersion compensation, and use linear receiver equalisation. Do not use soliton transmission or digital backpropagation.

Final engineering decision

YES – I would deploy this transmitter solution, subject to two conditions verified at commissioning.

Technical justification:

1. **Source:** $E_g = 0.80$ eV gives 1550 nm; $\eta_i = 80$ %; the DFB is single-mode with SMSR ≥ 35 dB, unlike LED/FP/VCSEL/QD.
2. **Bandwidth and power:** 20 GHz ≥ 17.5 GHz (2.5 GHz margin); 8.4 mW (9.24 dBm) covers all on-chip losses.
3. **Wavelength stability:** 4 nm drift is reduced to ±0.01 nm by a ±0.1 °C TEC, 10× inside the ±0.1 nm tolerance.
4. **Nonlinear impairments:** $\varphi_{NL} = 0.33$ rad $\ll$ 1 rad; FWM ≈ −45 dB; SRS margin 18 dB; SBS controlled by dither.
5. **Soliton/compensation:** N = 0.171, so the link is correctly designed as quasi-linear with optical dispersion management; DBP is not needed or possible.
6. **Standards and integration:** measurable compliance path for G.957/G.694.1/IEEE 802.3bs/bm; InP PIC gives the smallest loss, footprint and power for 8 channels.

Conditions: (a) SBS dithering must be confirmed by measuring back-reflected power at 2 mW launch; (b) APR shutdown must be tested on a simulated fiber break before the link is declared Hazard Level 1M compliant. If the bandwidth margin proves too small after packaging, the fallback is an EML (DFB + EAM) PIC or a higher-bandwidth MZM.

Assumptions

- Exponents in the PDF were unreadable; standard values used: $g_B = 5 \times 10^{-11}$ m/W, $g_R = 6 \times 10^{-14}$ m/W.
- T = 15 ps is taken as the soliton half-width T_0.
- D ≈ 15.7 ps/(nm·km) derived from $|\beta_2| = 20$ ps²/km at 1550 nm.
- FP example cavity: L = 300 µm, n = 3.5, R = 0.32, $\alpha_i = 10$ cm⁻¹. DML chirp estimate: $\alpha_H = 4$, 6 dB ER, 20 ps edges. Current-tuning rate 0.01 nm/mA.
- Losses: MZM 3 dB + 3 dB modulation, AWG 2.5 dB, fiber coupling 1.5 dB.
- Eye-safety limits at 1550 nm (approximate): Hazard Level 1 ≈ 10 mW, Hazard Level 1M ≈ 136 mW (ANSI Z136.2 / IEC 60825-2).
- SBS threshold increase: ×2 for NRZ carrier reduction, ×10 for ≈ 200 MHz phase dither.
- Ambient range 5–45 °C; TEC set-point 35 °C.
- ITU-T G.694.1 grid assumed in addition to G.957 for DWDM channel frequencies.